CS 506: Introduction to Quantum Computing

Simon's Algorithm

Scribe: Neeraj Vijayakumar Pattanashetti

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1 Overview

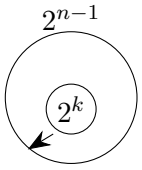
This lecture considers the probability of obtaining linearly independent vectors from $S^\perp$, the goal of uniformly sampling from $S^\perp$, and the Hadamard transformations used for paired inputs. It then gives the quantum oracle U_f and circuit, followed by the $n = 3$, $\vec{s} = 010$ XOR-pair and coset example.

2 Linear Independence in $S^\perp$

2.1 The Set and Its Size

$$S^\perp = \{\vec{z} \in \{0,1\}^n \mid \vec{z} \cdot \vec{s} = 0 \pmod 2\}$$

Has 2^{n-1} elements when $\vec{s} \neq \vec{0}$.



2.2 Probability of Linearly Independent Samples

For $n-1$ vectors sampled uniformly and independently from $S^\perp$:

$$\begin{aligned} P_n &= \Pr[\vec{z}_1, \vec{z}_2, \ldots, \vec{z}_{n-1} \text{ are l.i.}] \\ &= \left(1 - \frac{1}{2^{n-1}}\right)\left(1 - \frac{2}{2^{n-1}}\right)\left(1 - \frac{2^2}{2^{n-1}}\right)\cdots\left(1 - \frac{2^{n-2}}{2^{n-1}}\right). \end{aligned}$$

2.3 Product Limit and Lower Bound

$$\lim_{n\to\infty} P_n = \left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{2^2}\right)\left(1 - \frac{1}{2^3}\right)\cdots \approx 0.28879\ldots$$

$$\begin{aligned} P_n &= \frac{1}{2}\left(1 - \frac{1}{2^{n-1}}\right)\left(1 - \frac{2}{2^{n-1}}\right)\cdots\left(1 - \frac{2^{n-3}}{2^{n-1}}\right) \\ &\geq \frac{1}{2}\left(1 - \sum_{k=0}^{n-3} \frac{2^k}{2^{n-1}}\right). \end{aligned}$$

$$\sum_{k=0}^{n-3} \frac{2^k}{2^{n-1}} = \frac{1}{2^{n-1}} + \frac{1}{2^{n-2}} + \cdots + \frac{1}{2^2} < \sum_{j=2}^{\infty} \frac{1}{2^j} = \frac{\frac{1}{2^2}}{1 - \frac{1}{2}} = \frac{1}{2}.$$

$$\therefore \quad P_n > \frac{1}{2}\left(1 - \frac{1}{2}\right) = \frac{1}{4}.$$

3 Uniform Sampling from $S^\perp$

3.1 Set and Sampling Goal

$$S = \{\vec{0}, \vec{s}\}$$

$$S^\perp = \{\vec{z} \in \{0,1\}^n \mid \vec{z} \cdot \vec{s} = 0 \pmod 2\} \qquad \text{dot product}$$

Goal: design quantum circuit to output a string from $S^\perp$ uniformly at random.

3.2 Hadamard Transformation

We know:

$$\vec{x} = (x_1, x_2, \ldots, x_n) \in \{0,1\}^n$$

$$\begin{aligned} H^{\otimes n}|\vec{x}\rangle &= (H|x_1\rangle) \otimes (H|x_2\rangle) \otimes \cdots \otimes (H|x_n\rangle) \\ &= \frac{1}{\sqrt{2^n}} \sum_{\vec{z} \in \{0,1\}^n} (-1)^{\vec{x}\cdot\vec{z}} |\vec{z}\rangle \qquad \text{dot product modulo 2.} \end{aligned}$$

$$H^{\otimes n}|\vec{0}\rangle = \frac{1}{\sqrt{2^n}} \sum_{\vec{z} \in \{0,1\}^n} |\vec{z}\rangle$$

3.3 Equal Superposition of $|\vec{0}\rangle$ and $|\vec{s}\rangle$

$$\begin{aligned} & H^{\otimes n}\left(\frac{1}{\sqrt{2}}|\vec{0}\rangle + \frac{1}{\sqrt{2}}|\vec{s}\rangle\right) \\ &= \frac{1}{\sqrt{2}}\left(H^{\otimes n}|\vec{0}\rangle + H^{\otimes n}|\vec{s}\rangle\right) \\ &= \frac{1}{\sqrt{2}}\left[\frac{1}{\sqrt{2^n}} \sum_{\vec{z} \in \{0,1\}^n} |\vec{z}\rangle + \frac{1}{\sqrt{2^n}} \sum_{\vec{z} \in \{0,1\}^n} (-1)^{\vec{s}\cdot\vec{z}} |\vec{z}\rangle\right] \\ &= \frac{1}{\sqrt{2^{n+1}}}\left[\sum_{\vec{z} \in \{0,1\}^n} \left(1 + (-1)^{\vec{s}\cdot\vec{z}}\right) |\vec{z}\rangle\right] \end{aligned}$$

$$\begin{aligned} \vec{s}\cdot\vec{z} = 0 \quad &: \quad 1 + 1 = 2, \\ \vec{s}\cdot\vec{z} = 1 \quad &: \quad 1 - 1 = 0. \end{aligned}$$

4 Hadamard Transformation for Paired Inputs

4.1 Completing the Previous Transformation

$$\begin{aligned}&\frac{1}{\sqrt{2^{n+1}}}\sum_{\vec{z}\in S^{\perp}} 2|\vec{z}\rangle\\ &\quad=\frac{1}{\sqrt{2^{n-1}}}\sum_{\vec{z}\in S^{\perp}}|\vec{z}\rangle\end{aligned}$$

4.2 The XOR Relation

$$\vec{x},\vec{y}\in\{0,1\}^n,\qquad \vec{s}=\vec{x}\oplus\vec{y},\qquad \vec{x}\oplus\vec{s}=\vec{y}$$

$$\begin{aligned}s_1&=(x_1+y_1)\bmod 2,\\ s_2&=(x_2+y_2)\bmod 2,\\ &\vdots\\ s_n&=(x_n+y_n)\bmod 2.\end{aligned}$$

4.3 General Pair of Inputs

$$\begin{aligned}&H^{\otimes n}\left(\frac{1}{\sqrt{2}}|\vec{x}\rangle+\frac{1}{\sqrt{2}}|\vec{y}\rangle\right)\\ &=\frac{1}{\sqrt{2}}\left[H^{\otimes n}|\vec{x}\rangle+H^{\otimes n}|\vec{y}\rangle\right]\\ &=\frac{1}{\sqrt{2}}\left[\frac{1}{\sqrt{2^n}}\sum_{\vec{z}\in\{0,1\}^n}(-1)^{\vec{z}\cdot\vec{x}}|\vec{z}\rangle+\frac{1}{\sqrt{2^n}}\sum_{\vec{z}\in\{0,1\}^n}(-1)^{\vec{z}\cdot\vec{y}}|\vec{z}\rangle\right]\\ &=\frac{1}{\sqrt{2^{n+1}}}\left[\sum_{\vec{z}\in\{0,1\}^n}\left((-1)^{\vec{z}\cdot\vec{x}}+(-1)^{\vec{z}\cdot\vec{y}}\right)|\vec{z}\rangle\right].\end{aligned}$$

4.4 Amplitude Cases and Orthogonality

$$(-1)^{\vec{z}\cdot\vec{x}}+(-1)^{\vec{z}\cdot\vec{y}}=\begin{cases}-2, & (\vec{z}\cdot\vec{x},\,\vec{z}\cdot\vec{y})=(1,1),\\ +2, & (\vec{z}\cdot\vec{x},\,\vec{z}\cdot\vec{y})=(0,0),\\ 0, & (\vec{z}\cdot\vec{x},\,\vec{z}\cdot\vec{y})=(1,0)\text{ or }(0,1),\end{cases}$$

where the dot products are modulo 2. For the $(1,1)$ case:

$$\begin{aligned}z_1x_1+z_2x_2+\cdots+z_nx_n&=1 \pmod 2,\\ z_1y_1+z_2y_2+\cdots+z_ny_n&=1 \pmod 2,\\ z_1(x_1+y_1)+z_2(x_2+y_2)+\cdots+z_n(x_n+y_n)&=0 \pmod 2,\\ z_1s_1+z_2s_2+\cdots+z_ns_n&=0 \pmod 2,\\ \vec{z}\cdot\vec{s}&=0 \pmod 2,\quad \vec{z}\in S^{\perp}.\end{aligned}$$

The $(0,0)$ case likewise gives $\vec{z}\cdot\vec{s}=0 \pmod 2$.

5 Quantum Oracle and Circuit

5.1 Hadamard Result

$$\frac{1}{\sqrt{2^{n-1}}} \sum_{\vec{z} \in S^{\perp}} (-1)^{\vec{x} \cdot \vec{z}} |\vec{z}\rangle$$

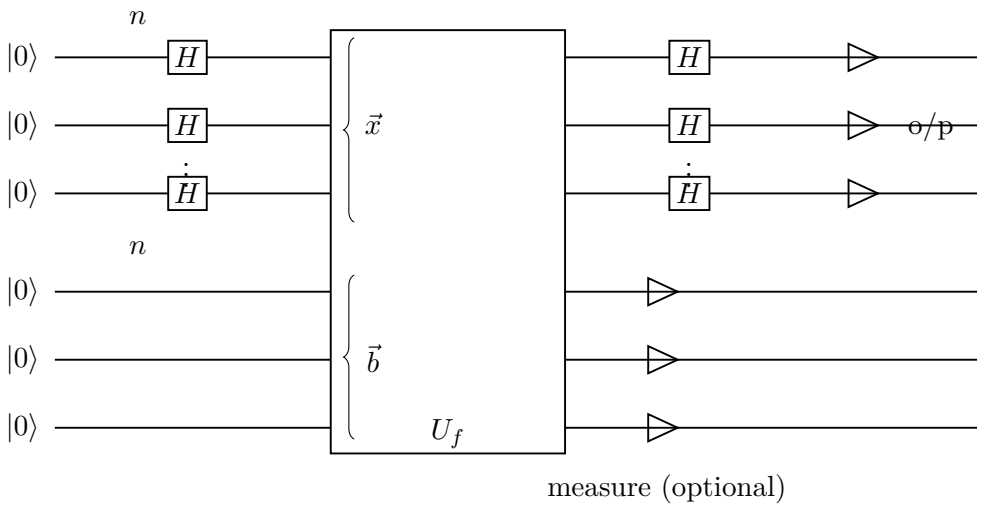
$$H^{\otimes n} \left[\frac{1}{\sqrt{2}} |\vec{x}\rangle + \frac{1}{\sqrt{2}} |\vec{x} \oplus \vec{s}\rangle \right] = \frac{1}{\sqrt{2^{n-1}}} \sum_{\vec{z} \in S^{\perp}} (-1)^{\vec{x} \cdot \vec{z}} |\vec{z}\rangle$$

5.2 Oracle U_f

$$f : \{0,1\}^n \longrightarrow \{0,1\}^n$$

$$U_f \left(|\vec{x}\rangle |\vec{b}\rangle \right) = |\vec{x}\rangle |\vec{b} \oplus f(\vec{x})\rangle, \qquad \vec{b} \in \{0,1\}^n$$

5.3 Quantum Circuit

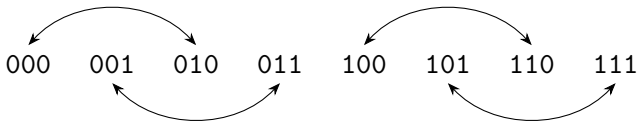


6 Example: $n = 3$, $\vec{s} = 010$

6.1 XOR Pairs and Equal Function Values

$$n = 3, \qquad \vec{s} = 010$$

$$\begin{aligned} 000 \oplus 010 &= 010, & f(000) &= f(010), \\ 001 \oplus 010 &= 011, & f(001) &= f(011), \\ 100 \oplus 010 &= 110, & f(100) &= f(110), \\ 101 \oplus 010 &= 111, & f(101) &= f(111). \end{aligned}$$



6.2 Cosets of the Subgroup

$$\{000, 010\}, \quad \{001, 011\}, \quad \{100, 110\}, \quad \{101, 111\}$$

Same value of f.

Cosets of subgroup $\{\vec{0}, \vec{s}\}$ over $\mathbb{Z}_2^n$

Inputs can be partitioned into groups of 2:

each group is of the form $\{\vec{x}, \vec{x} \oplus \vec{s}\}$.